When are the coefficients of $B(B(B(x)))$ odd?

A proof of two conjectures in OEIS A396093

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Abstract

Substituting the power series $B(x) = x/(1-x)^2 = x + 2x^2 + 3x^3 + \cdots$ into itself twice gives the series $A(x) = B(B(B(x))) = x + 6x^2 + 33x^3 + 174x^4 + \cdots$, whose coefficients form the sequence A396093 of the On-Line Encyclopedia of Integer Sequences (OEIS). Paul D. Hanna conjectured two rules for when these coefficients are even. We prove both, and more: the coefficient of x^n is odd if and only if n leaves remainder 1, 3, 7 or 9 on division by 10. The proof works with coefficients modulo 2, where the triple substitution collapses to the simple fraction $(x + x^3 + x^7 + x^9)/(1 + x^{10})$. The result has also been checked by computer and proved formally in the Lean proof assistant.

How this paper was produced. This paper reports the result of an autonomous AI research effort. ChatGPT (OpenAI), using the GPT-6 model with Pro reasoning and working from prompts by Dev D. Goyal, found the proof, wrote its Lean 4 formalization, and wrote a first draft of the paper. Dev D. Goyal chose the problems and asked for checks and revisions, but did not supply proof ideas. Claude Opus 5.5, used with Claude Code, then checked the work independently: it rebuilt the formal proofs from source and confirmed that they rely on no unproved assumptions, recomputed the sequence from its definition and tested every claim against the published data, and read every step of the proofs. It also rewrote this paper for clarity. Dev D. Goyal has verified the results and is responsible for them. The paper has not been peer reviewed. The section "About this paper" at the end gives the details.

1 Introduction

1.1 The sequence

The *On-Line Encyclopedia of Integer Sequences* (OEIS) [2] is a large public database of integer sequences, used by mathematicians and scientists to identify sequences and to record what is known about them. Many entries also record *conjectures*: patterns that the contributor observed in the data but could not prove.

This paper is about entry A396093 [1], contributed by Paul D. Hanna. It starts from the series

$$B(x) = \frac{x}{(1-x)^2} = x + 2x^2 + 3x^3 + 4x^4 + \cdots ,$$

whose coefficient of x^n is n. Substituting B into itself twice gives a new series,

$$A(x) = B(B(B(x))) = \sum_{n\geq 0} a(n)\, x^n = x + 6x^2 + 33x^3 + 174x^4 + 892x^5 + \cdots .$$

(Section 2 explains what substituting one series into another means.) The coefficients $a(n)$ form the sequence; its first terms, starting at $n = 0$, are 0, 1, 6, 33, 174, 892, 4480, 22149, 108144, 522685, 2505112, 11921919, ... The entry reads:

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G.f.  A(x) = B(B(B(x))), the third iteration of B(x) = x/(1-x)^2.

Conjecture:  a(2*n) is even for n >= 1.

Conjecture:  a(2*n-1) is even iff n is of the form 5*k-2 (k >= 1) for n >=
1.
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— OEIS A396093, Paul D. Hanna, May 2026

Here "G.f." stands for *generating function*, the series whose coefficients are the terms of the sequence.

1.2 The question

The two conjectures say when $a(n)$ is even. The first says that every term with an even index is even. The second says that a term with an odd index $2n-1$ is even only when n is one of $3, 8, 13, 18, \ldots$, that is, when the index is one of $5, 15, 25, 35, \ldots$. The table shows the parity of the first forty terms (1 means odd, 0 means even). The indices printed in bold are those that leave remainder 1, 3, 7 or 9 on division by 10.

n	0	**1**	2	**3**	4	5	6	**7**	8	**9**	10	**11**	12	**13**	14	15	16	**17**	18	**19**
$a(n)$ mod 2	0	**1**	0	**1**	0	0	0	**1**	0	**1**	0	**1**	0	**1**	0	0	0	**1**	0	**1**
n	20	**21**	22	**23**	24	25	26	**27**	28	**29**	30	**31**	32	**33**	34	35	36	**37**	38	**39**
$a(n)$ mod 2	0	**1**	0	**1**	0	0	0	**1**	0	**1**	0	**1**	0	**1**	0	0	0	**1**	0	**1**

The data suggest a cleaner statement, which is what we prove.

1.3 What this paper proves

Main result (Theorem 3.1). *The coefficient $a(n)$ is odd if and only if n leaves remainder* 1, 3, 7 *or* 9 *on division by* 10.

Both of Hanna's conjectures follow at once (Corollary 3.2). The key step is a short calculation showing that, once every coefficient is replaced by its remainder modulo 2, the series $A(x)$ becomes the simple fraction

$$\frac{x + x^3 + x^7 + x^9}{1 + x^{10}} = (x + x^3 + x^7 + x^9)\,(1 + x^{10} + x^{20} + \cdots),$$

from which the pattern can be read off directly.

1.4 What was known

The OEIS entry also records a closed formula for $A(x)$ as a ratio of two polynomials (due to Vaclav Kotesovec), and a linear recurrence for $a(n)$ (due to Seiichi Manyama). For any sequence given by such a ratio, the remainders modulo 2 are known to repeat periodically from some point on: there are only finitely many possible "states" of the recurrence modulo 2. So a periodic parity pattern was to be expected, and in principle its period could be found by a finite computation from the recorded formula. We do not rely on that formula. The proof below works directly from the definition and shows *why* the period is 10. We are not aware of an earlier proof of the two conjectures.

1.5 Why it is interesting

Substituting a series into itself is a basic operation: it describes what happens when a process is repeated, and iterates of simple series like B appear throughout combinatorics. Their coefficients quickly become large and irregular. Remainders modulo small numbers are often the first place where structure becomes visible, and explaining such patterns is a standard way of understanding a sequence. The example also shows a general technique: computing modulo 2 can collapse a complicated expression (here a ratio of two polynomials of degrees 7 and 8) into something transparent. Section 6 describes related questions.

1.6 How to read this paper

Section 2 explains the few notions we need: power series, substitution, and remainders. Section 3 states the results precisely, and Section 4 proves them. Section 5 describes how the results were checked, by computer and by a formal proof in Lean. Section 6 discusses the scope of the result and open questions.

2 Background

This section collects the definitions and facts used later. Readers who know formal power series and congruences can skip it and return when needed.

2.1 Formal power series

A *formal power series* is an expression

$$U(x) = u_0 + u_1 x + u_2 x^2 + u_3 x^3 + \cdots ,$$

determined by its sequence of coefficients $u_0, u_1, u_2, \ldots$. No number is ever substituted for x, so questions of convergence do not arise. We write $[x^n]\,U = u_n$ for the coefficient of x^n in U; two series are equal when all their coefficients are equal.

Series are added coefficient by coefficient and multiplied like polynomials:

$$[x^n]\,(UV) = u_0 v_n + u_1 v_{n-1} + \cdots + u_n v_0.$$

Each coefficient of a product is a finite sum. For example, $(1-x)(1+x+x^2+x^3+\cdots) = 1$, so $1+x+x^2+\cdots$ is the *multiplicative inverse* of $1-x$, written $\frac{1}{1-x}$. More generally, every series D with constant term 1 has a unique multiplicative inverse $\frac{1}{D}$, whose coefficients are determined one at a time by $D \cdot \frac{1}{D} = 1$ using only addition, subtraction and multiplication. Every fraction in this paper has a denominator of this kind, so it denotes a power series.

We write $U = O(x^d)$ when all coefficients of U below degree d are zero. So $U - V = O(x^d)$ means that U and V agree up to degree $d-1$. A series of the form $x + u_2 x^2 + u_3 x^3 + \cdots$, with constant term 0 and coefficient of x equal to 1, is called *normalized*.

2.2 Substituting one series into another

If $V(x)$ has constant term 0, we can substitute V for x in U:

$$U(V(x)) = u_0 + u_1 V(x) + u_2 V(x)^2 + u_3 V(x)^3 + \cdots .$$

This is called *composition*, also written $U \circ V$. It makes sense because $V(x)^k$ has no terms below x^k, so only the finitely many terms with $k \le n$ affect the coefficient of x^n. For example, if $U = x + x^2$ and $V = x^2$, then $U(V(x)) = x^2 + x^4$. Composition is associative: $U \circ (V \circ W) = (U \circ V) \circ W$.

Substituting a series into itself repeatedly gives its *iterates*: $U^{\circ 2}(x) = U(U(x))$, $U^{\circ 3}(x) = U(U(U(x)))$, and so on, with $U^{\circ 0}(x) = x$ and $U^{\circ 1} = U$. They must not be confused with *powers* $U^2 = U \cdot U$, $U^3 = U \cdot U \cdot U$. (The OEIS sometimes writes $A^k(x)$ for the k-th iterate. In this paper the small circle always means an iterate, and a plain exponent always means a power.)

2.3 Remainders and congruences

For an integer $m \ge 2$ we write $a \equiv b \pmod m$ ("a is congruent to b modulo m") when $a - b$ is divisible by m, that is, when a and b leave the same remainder after division by m. For example, $17 \equiv 5 \pmod 4$, since both leave remainder 1, and $-3 \equiv 1 \pmod 4$. The remainder of a on division by m is called its *residue* modulo m; it is one of $0, 1, \dots, m-1$. Saying that a is odd is the same as saying $a \equiv 1 \pmod 2$.

Addition and multiplication are compatible with congruences, so the residues $0, 1, \dots, m-1$, added and multiplied modulo m, form a ring, written $\mathbb{Z}/m\mathbb{Z}$. When $m = p$ is a prime we also write $\mathbb{F}_p$. In $\mathbb{F}_2 = \{0, 1\}$ we have $1 + 1 = 0$, so addition and subtraction coincide.

For power series with integer coefficients, $U \equiv V \pmod m$ means that *every* coefficient of $U - V$ is divisible by m. Equivalently, the two series become equal when every coefficient is replaced by its residue; we call this *reducing* the series modulo m. The key fact is the following.

Reduction modulo m is compatible with all the operations above. To reduce a sum, a product, a substitution, or an inverse, we may reduce the ingredients first and then compute in $\mathbb{Z}/m\mathbb{Z}$. This holds because every coefficient of the result is obtained from finitely many coefficients of the ingredients by adding, subtracting and multiplying, and these operations are compatible with remainders.

2.4 Squaring modulo 2

Lemma 2.1 (Frobenius identity)**.** *Let p be a prime. For every power series U with coefficients in $\mathbb{F}_p$, we have $U(x)^p = U(x^p)$.*

Proof. In $\mathbb{F}_p$, the binomial theorem simplifies to $(a + b)^p = a^p + b^p$, because every binomial coefficient $\binom{p}{i}$ with $0 < i < p$ is divisible by p. Also $c^p = c$ for every $c \in \mathbb{F}_p$ (Fermat's little theorem). So $\left(\sum_n u_n x^n\right)^p = \sum_n u_n^p x^{pn} = \sum_n u_n x^{pn} = U(x^p)$. Only finitely many terms matter for any given coefficient, so this is a polynomial identity. □

For $p = 2$ this says that squaring a series modulo 2 doubles every exponent: for instance, $(x + x^3)^2 \equiv x^2 + x^6 \pmod 2$. In particular, the square of any series has, modulo 2, no terms of odd degree.

3 Results

Theorem 3.1. *For every $n \ge 0$,*

$$a(n) \text{ is odd} \quad \Longleftrightarrow \quad n \equiv 1,\ 3,\ 7 \text{ or } 9 \pmod{10}.$$

Equivalently, modulo 2 the generating function is

$$A(x) \equiv \frac{x + x^3 + x^7 + x^9}{1 + x^{10}} = \sum_{j \geq 0} (x^{10j+1} + x^{10j+3} + x^{10j+7} + x^{10j+9}) \pmod 2. \tag{1}$$

Corollary 3.2 (Hanna's conjectures)**.** *(a) $a(2n)$ is even for every $n \geq 0$.*

(b) For $n \geq 1$, $a(2n-1)$ is even if and only if $n = 5k - 2$ for some integer $k \geq 1$.

Proof. (a) An even number never leaves remainder 1, 3, 7 or 9 on division by 10. (b) An odd number $2n - 1$ leaves one of the remainders $1, 3, 5, 7, 9$. By Theorem 3.1, $a(2n-1)$ is even exactly in the remaining case $2n - 1 \equiv 5 \pmod{10}$. This means $2n \equiv 6 \pmod{10}$, that is, $n \equiv 3 \pmod 5$. For $n \geq 1$ these are the numbers $3, 8, 13, \ldots$, which are exactly the numbers $5k - 2$ with $k \geq 1$. □

4 Proofs

The idea in brief. We reduce all coefficients modulo 2. There, $B(x) = x/(1 + x^2)$, and substituting a fraction into B gives another fraction by a simple rule (Lemma 4.1). Applying the rule three times gives A as a fraction, and multiplying top and bottom by $1 + x^2$ turns the denominator into $1 + x^{10}$.

From now on we work in $\mathbb{F}_2[[x]]$, that is, with coefficients modulo 2, and we keep the names A and B for the reduced series. Because reduction is compatible with substitution, the reduction of $A = B(B(B(x)))$ equals the triple substitution of the reduction of B. Since $-1 = 1$ in $\mathbb{F}_2$, and since $(1 + x)^2 = 1 + x^2$ there (Lemma 2.1),

$$B(x) = \frac{x}{(1 - x)^2} = \frac{x}{(1 + x)^2} = \frac{x}{1 + x^2}.$$

Lemma 4.1. *Let P and D be series over $\mathbb{F}_2$ with $P(0) = 0$ and $D(0) = 1$. Then*

$$B\left(\frac{P}{D}\right) = \frac{P\,D}{D^2 + P^2}.$$

Proof. Put $T = P/D$. Then $B(T) = T/(1 + T^2)$. Multiplying the numerator and the denominator by D^2 gives $PD/(D^2 + P^2)$. The new denominator has constant term 1, so this fraction is again a power series. □

Proof of Theorem 3.1. Starting from $x = x/1$ and applying Lemma 4.1 three times (all arithmetic modulo 2):

series	numerator P	denominator D
$B(x)$	x	$1 + x^2$
$B(B(x))$	$x(1 + x^2) = x + x^3$	$(1 + x^2)^2 + x^2 = 1 + x^2 + x^4$
$B(B(B(x)))$	$(x + x^3)(1 + x^2 + x^4) = x + x^7$	$(1 + x^2 + x^4)^2 + (x + x^3)^2 = 1 + x^2 + x^4 + x^6 + x^8$

For instance, in the last row $(x + x^3)(1 + x^2 + x^4) = x + 2x^3 + 2x^5 + x^7$, and the terms with coefficient 2 vanish modulo 2. Hence

$$A(x) = \frac{x + x^7}{1 + x^2 + x^4 + x^6 + x^8}.$$

Now multiply the numerator and the denominator by $1 + x^2$. The numerator becomes $(x+x^7)(1+x^2) = x+x^3+x^7+x^9$. The denominator becomes $(1+x^2)(1+x^2+x^4+x^6+x^8) = 1 + x^{10}$, because every middle term appears twice and so vanishes. This proves the first form of (1). Finally, $1/(1 + x^{10}) = 1 + x^{10} + x^{20} + \cdots$ (check by multiplying out), which gives the second form. Reading off coefficients, $a(n)$ is odd exactly when $n \equiv 1, 3, 7, 9 \pmod{10}$. □

5 Checking the results

A mathematical proof can contain mistakes, so the results were checked in two independent ways.

Numerical check. We computed $B(B(B(x)))$ directly from the definition. The result reproduces all 27 terms printed in the OEIS entry, and the parity rule of Theorem 3.1 holds for every $n < 600$. The computation was done with the PARI/GP computer algebra system, using code that was written separately from the formal proofs (file `verification/independent_checks.gp` in the accompanying repository [7]). A computation can only test finitely many cases, so it is a check on the definitions and on the indexing, not a proof.

Formal proof in Lean. Lean [5] is a *proof assistant*: a program in which mathematical statements and their proofs are written in a precise formal language, and every step is checked mechanically down to the basic rules of logic. All results of this paper are proved in Lean 4 (version 4.19.0) on top of Mathlib [6], Lean's library of formalized mathematics. The statements are collected in the file `lean/Entries/A396093.lean` of the repository, in the namespace `OEIS.A396093`. There B is defined by the equation $B \cdot (1 - x)^2 = x$ and A as the actual threefold composition. The table lists where each result of this paper is proved.

Statement in this paper	Lean theorem(s)
$B(x) \cdot (1 - x)^2 = x$ (definition of B)	`source_denominator`
$A = B \circ B \circ B$, and $a(n)$ is its coefficient of x^n	`source_composition`, `coefficient_definition`
Equation (1): $A \cdot (1 + x^{10}) = x + x^3 + x^7 + x^9$ over $\mathbb{F}_2$	`reduced_denominator`
Theorem 3.1	`complete_parity`, `even_iff`
Corollary 3.2(a) and (b)	`even_indices`, `odd_indices`

The Lean project was rebuilt from source for this paper, and Lean was asked to list the basic assumptions (axioms) that each of these theorems ultimately depends on. Every one depends only on the three standard axioms of Lean's logic (`propext`, `Classical.choice`, `Quot.sound`); no step is left unproved. A formal proof certifies exactly the statement that Lean checks. Whether that statement says what the OEIS entry means is a separate question, so the Lean definitions were also compared, side by side, with the definition in the OEIS entry (Section 1.1).

6 Discussion

Scope. Theorem 3.1 describes the parity of every coefficient, so it settles both conjectures of the entry completely. It says nothing about the remainders of $a(n)$ modulo other numbers, or about the size of $a(n)$.

Other iterates. The same calculation applies to every iterate of B. The first two rows of the table above already show that $B(x) \equiv x/(1+x^2)$ and, after multiplying by $1+x^2$, that $B(B(x)) \equiv (x+x^5)/(1+x^6) \pmod 2$. So the coefficients of $B(B(x))$ (OEIS A030267) are odd exactly when $n \equiv 1$ or $5 \pmod 6$. For the fourth iterate, a computer calculation shows a parity pattern of period 34; we did not find a simple rule that predicts the period for every iterate, and this seems a natural question.

Higher moduli. A computer calculation suggests that $a(n)$ modulo 4 is also periodic, with period 20, starting from $n = 0$. Since A is a ratio of polynomials, some period is guaranteed; the method of this paper could be used to determine it exactly.

About this paper

How the work was done. This paper is one of 26 papers on conjectures from the OEIS that came out of an autonomous AI research effort directed by Dev D. Goyal [7]. The roles were as follows.

- *Dev D. Goyal* started the project, chose which problems to pursue, asked for proofs, formal verification and revisions, and decided what to publish. Dev D. Goyal did not supply proof strategies or intermediate arguments.
- *ChatGPT* (OpenAI), using the GPT-6 model with Pro reasoning, worked autonomously from those prompts. It searched the OEIS for suitable conjectures, found the proofs, wrote the Lean 4 formalization, and wrote a first draft of this paper.
- *Claude Opus 5.5*, used with Claude Code, checked the work independently and then rewrote this paper. It rebuilt the Lean project from source and confirmed that the theorems listed in Section 5 rely only on Lean's standard axioms; compared the Lean definitions with the OEIS definition; recomputed the sequence with separately written PARI/GP code, compared it with the terms published in the OEIS, and tested every claim of the paper on many terms; and read every step of the proofs. It then rewrote the text to make it self-contained and readable. The theorems and the proof strategies are ChatGPT's. In rewriting, Claude Opus 5.5 expanded the explanations and, in places, wrote out an argument in more detail or by a more elementary route. The Discussion section was added by Claude Opus 5.5; the further observations in it were checked by computer and are not part of the Lean formalization.

Dev D. Goyal has verified the results and is responsible for this paper.

Review status. This paper has not been peer reviewed, and it has not been submitted to a journal or to the OEIS.

Priority. An OEIS entry that labels a statement as a conjecture shows only that its contributor did not know a proof; it does not show that no proof existed. During the research only limited searches for earlier work were made. No claim of priority is made.

Corrections. Please report any error in the issue tracker of the repository [7], where the Lean sources and the numerical checks are also available.

Acknowledgements

The sequence and both conjectures are due to Paul D. Hanna [1]. The closed formula and recurrence mentioned in the introduction were contributed to the entry by Vaclav Kotesovec and Seiichi Manyama.

References

[1] P. D. Hanna, entry A396093 (Third iteration of $x/(1-x)^2$), *The On-Line Encyclopedia of Integer Sequences*, May 2026. `https://oeis.org/A396093`. Text as of October 2026.

[2] N. J. A. Sloane and The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*. `https://oeis.org`.

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[4] I. Niven, Formal power series, *American Mathematical Monthly* 76 (1969), 871–889.

[5] L. de Moura and S. Ullrich, The Lean 4 theorem prover and programming language, in *Automated Deduction – CADE 28*, Lecture Notes in Computer Science 12699, Springer, 2021, 625–635.

[6] The mathlib Community, The Lean mathematical library, in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, 367–381. Version used: commit `c44e0c8ee63c`.

[7] D. D. Goyal, *Proofs and corrections of OEIS conjectures, with Lean 4 formalizations*, repository, 2026. `https://github.com/dvgyl/oeis-a396843-proof`.